Systems Biology & Network Modeling (CB 5313)

Course Project — Addendum to the Syllabus

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Purpose

Each student takes one method from current network-biology research that the course does not cover, reads its primary paper, and applies the method to real data. Topics may be chosen by more than one student; all topics share the same deliverables and the same grading criteria.

Part 1 — Reading

1. **Annotated PDF** of the primary paper: highlights and margin notes showing what you understood, what you questioned, and how the method works.
2. **Synopsis**, in your own words, one page (400–600 words), with four headed sections:
   - *Question* — what biological or methodological problem the paper addresses.
   - *Method* — how it works, including its key assumptions.
   - *Key result* — the main finding and the evidence for it.
   - *Limitation* — at least one weakness, assumption, or failure mode, and why it matters.

Part 2 — Implementation

Run the method on a dataset you find in the tool's resources or online, and **make at least one change beyond the tool's tutorial**, for example:

- a different dataset from the one used in the tutorial;
- a parameter sweep, with its effect on the result;
- a comparison with a method taught in this course.

Interpret the outcome of the change; running a tutorial unchanged does not meet this requirement.

Submit a **GitHub repository** containing:

- well-commented code;
- a `README` stating the question, the data source, and how to run the code;
- an environment file (`requirements.txt`, `environment.yml`, or `renv.lock`);
- a script that downloads the data (do not commit large raw data);
- a results section that interprets the output, not only displays it.

Part 3 — Presentation

Present your project in class: about **10 minutes**, followed by **5 minutes of questions**. Cover:

- the question the paper addresses, and how the method works;
- the method's key assumptions and at least one limitation;
- the change you made beyond the tutorial, and why you chose it;
- your results and your interpretation of them.

Every student is asked questions of the following kinds:

1. **Purpose** — What question does this method answer that the methods taught in this course cannot?

2. **Assumptions** — Name one assumption of the method. What happens to the result if it is violated?
3. **Code** — Walk through a step of your code: what goes in, and what comes out?
4. **Change** — What did you expect before making your change? Why did the result change, or not change?
5. **What-if** — If you had more samples, different data, or a different threshold, what would change?
6. **Trust** — How would you know if your result is wrong?

You are expected to explain and defend everything you submitted, regardless of the tools you used to produce it.

Topics

All tools run on a standard laptop CPU; none require a GPU or MATLAB.

#	Topic	Primary paper	Tool (language)
1	Single-cell GRN inference	Aibar, S., et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11), 1083–1086.	pySCENIC (Python) / SCENIC (R)
2	Cell–cell communication	Jin, S., et al. (2021). Inference and analysis of cell-cell communication using CellChat. *Nature Communications*, 12, 1088.	CellChat (R)
3	Sample-specific networks	Kuijjer, M.L., et al. (2019). Estimating sample-specific regulatory networks. *iScience*, 14, 226–240.	LIONESS (netZooR / netZooPy)
4	Graph embeddings	Grover, A., & Leskovec, J. (2016). node2vec: scalable feature learning for networks. *Proc. KDD*, 855–864.	node2vec (Python)
5	Binary interactome mapping	Luck, K., et al. (2020). A reference map of the human binary protein interactome. *Nature*, 580, 402–408.	HuRI data + igraph (R / Python)
6	Directed GRN inference	Huynh-Thu, V.A., et al. (2010). Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*, 5(9), e12776.	GENIE3 (R / Python)
7	Context-specific metabolic models	Shlomi, T., et al. (2008). Network-based prediction of human tissue-specific metabolism. *Nature Biotechnology*, 26, 1003–1010.	iMAT via troppo or imatpy (Python); requires a MILP solver
8	Disease module detection	Ghiassian, S.D., Menche, J., & Barabási, A.-L. (2015). A DIseAse MOdule Detection (DIAMOnD) algorithm. *PLoS Computational Biology*, 11(4), e1004120.	DIAMOnD (Python)
9	Network-based drug repurposing	Guney, E., et al. (2016). Network-based in silico drug efficacy screening. *Nature Communications*, 7, 10331.	NetMedPy (Python)

#	Topic	Primary paper	Tool (language)
10	Network controllability	Liu, Y.-Y., Slotine, J.-J., & Barabási, A.-L. (2011). Controllability of complex networks. *Nature*, 473, 167–173.	Maximum matching (igraph / networkx)

Policies

Individual work. Discussion of concepts and tools is encouraged; the annotated PDF, synopsis, code, and interpretation must be your own. Students on the same topic must use different datasets or different changes.

AI use. AI assistance must be disclosed and cited, per the course policy. The synopsis and the biological interpretation must be your own.

Deadlines and weighting. To be announced.